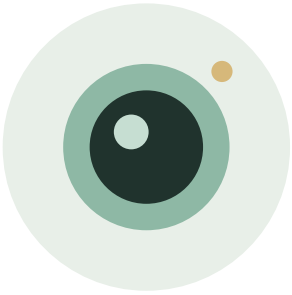
GODCAM

Capture More Than Moments.



DETAILED PRODUCT REQUIREMENTS DOCUMENT

A premium, mobile-first camera and film-look app built around effortless capture, powerful manual controls and honest high-quality export.

Visual direction: bold Apple-inspired clarity, warm cream surfaces, subtle mint-green accents, tactile controls and photography-first layouts. No generic AI-looking gradients or clutter.

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1. Executive Summary

GODCAM combines a responsive camera viewfinder, 100+ film-inspired looks, manual photo controls, advanced video recording and a built-in gallery with export tools. The default experience should be simple enough for quick snapshots while advanced features remain discoverable for enthusiasts.

Core promise: real capture, real processing and real export—not simulated controls. Resolution, FPS, bitrates, RAW, manual settings and 4K/8K output must be offered only when the device and processing pipeline genuinely support them.

Product goals

- Capture a beautiful photo or video quickly with excellent defaults.
- Offer 100+ distinct presets across Fuji-inspired, Kodak-inspired, retro, cinematic, monochrome, portrait, nature and experimental families.
- Provide meaningful manual controls without overwhelming casual users.
- Enable reliable local browsing, preview, editing, sharing and export.
- Make every screen adaptive to compact and large phones, orientation and system text size.

Non-goals

- Do not promise 8K or every frame rate on every phone.
- Do not fake progress, camera settings, file saves or exports.
- Do not require an account or internet for basic local capture.
- Do not request broad file access when Android Photo Picker or scoped storage is sufficient.

2. Audience and Core Use Cases

Audience	Need	Product response
Everyday creators	Great photos without complex editing	Auto camera, live looks, simple capture and export
Film-look enthusiasts	Consistent analog-inspired color	Filter families, intensity, favorites and comparison
Mobile videographers	Manual video capture	Pro Video controls, profile validation and quality export
Travel/night photographers	Low-light and scene capture	Night mode where supported, exposure and timer
Power users	Control over output	Custom dimensions, codec, FPS, bitrate and metadata controls

3. Platform and Technical Constraints

Recommended implementation: native Android with Kotlin and Jetpack Compose. Use CameraX for compatibility and Camera2 interop for advanced hardware controls. Use supported Android media/encoder APIs for video and processing, MediaStore for app-created media, and Android Photo Picker for user-selected imports.

Area	Direction	Must-have behavior
UI	Compose + reusable design system	Responsive layouts, accessible controls, smooth state transitions
Camera	CameraX / Camera2 interop	Runtime capability discovery; valid capture and manual control states
Video	Supported camera and encoder APIs	Filter profile combinations by actual hardware capability
Image pipeline	Efficient GPU/shader processing where practical	Consistent preview and saved-image rendering
Storage	MediaStore and app-private working files	Save, reopen, share, delete and export real files

Area	Direction	Must-have behavior
Permissions	Contextual requests	Camera; microphone for audio video; media picker; optional notifications
Cloud	Not required for core app	Local capture works offline

A device may not support 8K, high-FPS 4K, RAW, manual focus, manual shutter or all bitrates. Build a capability matrix at runtime and show only valid combinations. Never present unsupported options as working features.

4. Information Architecture

Destination	Purpose	Key features
Camera	Default capture screen	Photo/video, shutter/record, lens switch, flash, grid, EV, ratio, timer
Looks	Browse and apply presets	Search, categories, favorites, intensity, compare
Pro	Advanced capture	ISO, shutter, focus, white balance, metering and locks when supported
Gallery	Browse local media	Photos, videos, favorites, viewer, details, edit, share, export
Settings	Personalize app	Themes, defaults, storage, haptics, permissions and help

The camera preview is the hero. On compact phones, advanced tools should open in scrollable bottom sheets rather than squeeze the preview. Keep primary shutter/record controls reachable by thumb.

5. Visual Design System

Brand direction

- Default palette: warm cream background, deep forest-charcoal text, soft mint active states, muted sage borders and neutral surfaces.
- Use clear system typography, confident headings, tabular numerals for camera values and simple vector icons.
- Photography should be the main visual content. Avoid decorative blobs, excessive gradients, neon glow, glass layers and random ornament.
- Use purposeful motion for shutter feedback, focus confirmation and sheet transitions; respect reduced-motion preferences.

Token	Default	Use
Background	#F7F3E9	Main surfaces
Accent	#8EB8A5	Selected controls, toggles and capture accents
Primary text	#20332D	Headings, labels and important values
Secondary text	#66736C	Descriptions and metadata
Surface	#FCFBF7	Cards, fields and sheets
Border	#D9DED5	Quiet separators
Shape language	Pills + rounded cards	Mode selectors, chips, actions, sheets and fields
Icon style	Simple vector icons	Consistent stroke, meaningful labels

Component rules

- Primary and secondary actions use pill-shaped buttons; selected states use mint fill with dark readable text.
- Icon controls use circular touch targets with pressed, selected, disabled and accessibility states.
- Sliders show value and unit; provide fine adjustment and a one-tap reset where relevant.

- Cards and sheets have consistent rounded corners and alignment—never randomly different radii.
- The app icon artwork is a circular camera/lens emblem with no rectangle inside the central icon design. Android launchers may apply their own outer mask.

6. Main Camera Screen

- Full-bleed live viewfinder with safe-area-aware overlays.
- Top controls: flash/torch, settings, lens/camera selector and current capture profile when space allows.
- Preview overlays: optional grid, level, focus/exposure target, zoom/lens choices and EV value.
- Bottom controls: recent-media thumbnail, large shutter/record button, front/rear switch and mode selector.
- Quick tools: filters, aspect ratio, timer, grid, flash, exposure and resolution shortcuts.
- A single tap captures. Long-press quick video is optional and must not conflict with selected mode.
- Hide rarely used tools on small screens; never clip controls or cover key parts of the preview.

Photo controls

Feature	Requirements
Front/rear camera	Switch available cameras; preserve valid settings and clearly reset unsupported ones.
Flash/torch	Photo flash modes and continuous torch where hardware supports them.
Grid	Off, 3×3, 4×4 and optional golden-ratio overlay; not burned into image by default.
Exposure	EV slider with numeric value and device-supported range.
Focus	Tap-to-focus and focus lock; manual slider only when supported.
White balance	Auto and supported presets; Kelvin/tint only if reliable.
Zoom	Available physical lenses and digital zoom with selected lens identified.
Timer	Off, 3s, 5s, 10s; visible countdown and cancel.
Aspect ratios	4:3, 3:2, 16:9, 1:1, 4:5, 9:16; explain crop vs sensor capture.
Photo formats	JPEG baseline; HEIF/HEIC and RAW/DNG only when supported.
Resolution	Native sensor choices and custom export dimensions; distinguish capture resolution from export size.

Night mode

Night mode should detect low light or allow manual activation. Where supported, use multi-frame capture, alignment and noise reduction; otherwise provide a clearly labeled low-light path. Show progress for longer captures, prevent accidental interruption and do not imply the torch is computational night mode.

Capture feedback

- Visible focus indicator and optional AE/AF lock.
- Exposure compensation responds immediately and has a neutral reset.
- Clear states for storage full, overheating, camera busy, permission denied and unsupported lens.
- Optional haptics for shutter/recording, with a setting to disable them.

7. Filter Library: 100+ Looks

Launch with at least 100 visually distinct, offline-ready presets. Names should be original/descriptive; do not imply an official partnership or exact film-stock emulation without legal clearance.

Family	Look direction	Adjustments
Fuji-inspired	Soft greens, airy portraits, cinematic tones	Intensity, warmth, contrast, grain
Kodak-inspired	Warm highlights, nostalgic color, rich skin tones	Intensity, warmth, fade, grain
Retro/vintage	Faded color, analog warmth, optional edge effects	Intensity, fade, vignette, grain
Cinematic	Muted palettes and filmic contrast	Intensity, contrast, highlights
Monochrome	Classic B&W;, silver, noir, high contrast	Intensity, contrast, grain
Instant-film inspired	Soft contrast, warm whites, nostalgic rendering	Intensity, fade, optional border
Nature	Foliage, landscape, ocean and sky	Intensity, saturation, tone
Portrait	Skin-friendly color and controlled highlights	Intensity, warmth, contrast
Night/neon	City glow and restrained saturation	Intensity, tint, contrast
Creative	Duotone, cross-process-inspired experiments	Intensity and supported color controls

Filter behavior

- Every preset has a unique ID, name, family, thumbnail and rendering parameters.
- Live preview where performance allows; otherwise show an accurate sample and apply to captured media.
- Intensity from 0–100%, Original/Before comparison, search, category chips, favorites and recents.
- Preserve original captures; filtered outputs create a new copy unless replacement is explicitly requested.
- Use non-destructive edit recipes where practical so a look can be changed later.
- Test all presets across varied skin tones, lighting, landscapes and night scenes; avoid duplicate looks with different names.

8. Pro Photo Mode

Control	Requirement
ISO	Auto and supported manual range; show numeric value and Auto reset.
Shutter speed	Auto and valid exposure-time choices; show actual applied value where available.
Focus	Auto/manual focus only where supported, with clear near/far feedback.
White balance	Auto and supported presets; temperature/tint only when reliable.
Exposure compensation	EV slider using device-supported range.
Metering	Matrix/center/spot only when exposed by the hardware/API.
AE/AF lock	Separate or combined lock with clear active state.
Histogram/clipping	Optional overlay based on real preview/image data.
Capture format	JPEG, supported HEIF and genuine RAW/DNG capture when available.
Burst/timer	Only where supported without destabilizing manual capture.

Show actual camera state when the API exposes it. If a requested setting is approximated or ignored, say so. Never expose manual controls that the active camera path cannot reliably implement.

9. Video and Pro Video

Standard video

- Dedicated Video mode with large record control, elapsed timer, audio status and storage estimate.

- Offer supported resolutions such as 720p, 1080p, 4K or 8K only when available on the selected camera.
- Offer 24, 25, 30, 50, 60, 120 or 240 FPS only when supported by the selected resolution, camera and encoder.
- Show target bitrate/quality clearly; distinguish target from measured/actual output.
- Allow silent video if microphone permission is declined where feasible.
- Prevent invalid camera/lens changes during recording and handle interruptions safely.

Pro Video controls

Control group	Detailed requirements
Resolution	Supported dimensions; distinguish UHD 4K and DCI 4K when applicable; 8K only if supported.
Frame rate	Valid FPS choices filtered by camera, resolution, codec and device capability.
Bitrate	Auto plus supported manual target values or quality tiers; validate limits and free space.
Codec/format	MP4/H.264 baseline; HEVC/H.265 and others only when supported.
Shutter	Manual shutter speed/angle where supported; optional 180-degree guidance.
ISO/exposure	Manual/auto ISO and EV when supported; actual values and lock states.
White balance	Auto/manual presets or temperature where reliably supported.
Focus	Continuous AF, tap focus, focus lock and manual focus where available.
Stabilization	Expose available modes and explain crop/FPS trade-offs.
Audio	Mic status/source and level where supported; clear mute state.
Aspect ratio	16:9, 9:16, 1:1 and cinematic guides 2.35:1/2.39:1; label guides vs crops.
Color profile	Standard or flat/log-style only if truly supported; don't mislabel processed footage as log.
Assist overlays	Optional focus peaking/zebra if implemented; don't burn into footage by default.
Recording limits	Show storage, thermal, duration or file-size constraints when known.

Valid profile logic

Maintain a runtime matrix of valid combinations. Selecting a resolution filters FPS; selecting FPS filters resolution/codec; codec selection updates valid bitrate ranges. Unsupported requests should produce a clear explanation and nearest supported alternative. Warn about heat and storage pressure without silently downgrading quality.

Cinematic framing

- Offer 16:9, 2.00:1, 2.20:1, 2.35:1 and 2.39:1 guides where practical.
- Clearly label whether a ratio is a viewfinder guide, an export crop or native recording dimensions.
- Preserve original footage; create a separate cropped export when requested.
- Include vertical 9:16 and square 1:1 outputs for flexible sharing.

10. Gallery and Media Viewer

- Local Photos, Videos, Favorites and Albums/Collections views.
- Adaptive thumbnail grid, progressive loading and smooth scrolling.
- Show capture date, dimensions, duration and file size in details; avoid excessive badges.
- Use MediaStore for saved content and Photo Picker for user-selected external media.
- Support multi-select, favorite, delete with confirmation, share and export.
- Full-screen photos support pinch/double-tap zoom and swipe navigation.

- Video player includes play/pause, seek, duration, mute/volume and orientation handling.
- Actions include Edit, Apply Look, Compare Original, Details, Share and Export.
- Preserve original files; edits and exports create new files by default.

11. Export Studio

Export Studio must create actual output files and provide meaningful controls, preview, estimated size when possible, progress, cancellation and failure recovery.

Option	Requirements
Photo dimensions	Original, common presets and custom width/height; maintain ratio by default.
Photo format	JPEG baseline; HEIF/HEIC or PNG where appropriate; RAW only for genuine RAW capture.
Photo quality	Quality slider/presets, estimated file size and metadata options.
Video resolution	Original, supported presets and custom dimensions within codec limits.
Video FPS	Original or supported export FPS; frame conversion/interpolation only when real.
Bitrate/quality	Auto, named quality presets and manual target bitrate where supported.
Format/codec	MP4/H.264 baseline; HEVC/H.265 or others where available.
Aspect ratio	Original, common formats, cinematic crops and custom ratio with crop preview.
Audio	Keep or mute; supported audio quality controls without unexpected changes.
Metadata	Optional preserve/remove location and metadata with privacy explanation.
Destination	Save to gallery, save copy or Android share sheet; confirm actual result.
Progress/cancel	Real progress, cancel, background handling and interrupted-job recovery.

Custom dimensions

Provide sliders plus editable numeric width and height fields, aspect-ratio lock, reset and supported bounds. Respect encoder/decoder constraints, including even-dimension requirements where relevant. Never upscale silently.

4K/8K and quality honesty

4K/8K export is available only if source, pipeline, memory, encoder and device can complete it. Upscaling increases pixel dimensions but does not restore missing detail; label it clearly and make it optional. Validate hardware support before enabling expensive exports.

Preset	Purpose	Behavior
Efficient	Smaller shareable files	Lower bitrate/quality within sensible limits
Balanced	Everyday use	Good quality and moderate size
High	Best practical quality	Higher quality and larger output
Maximum/Original	Preservation	Original stream or highest valid output; no unsupported promises
Custom	Power users	Manual dimensions, codec, FPS and bitrate within valid limits

12. Themes and Personalization

Theme	Direction
Cream Mint (default)	Warm cream, subtle mint, deep green-black
Paper Sage	Paper white and muted sage
Ocean	Pale blue and restrained teal
Lavender Mist	Quiet lavender and neutral gray
Sunset Sand	Warm sand and muted terracotta
Graphite	Dark graphite with mint accents
Midnight	Deep dark surfaces and accessible contrast
System	Follow Android light/dark preference

- Theme swatches are circular and accessible; persist selection across launches.
- Support light, dark and system modes; never force dark mode.
- Respect system font scale and contrast; don't use mint for tiny text on cream.
- Themes alter colors and states, not the app's core structure.

13. Adaptive UX and Accessibility

- Design compact phones first, then scale to larger phones and foldables without stretching controls.
- Use adaptive window sizing; account for cutouts, system bars, keyboard, portrait and landscape.
- Keep shutter/record reachable by thumb; advanced settings use scrollable sheets.
- No horizontal overflow, clipped sliders, tiny targets or fixed-size layouts.
- Support screen readers, large fonts, meaningful control labels, values and states.
- Use non-color-only status cues and accessible contrast.
- Test camera preview orientation and media metadata after rotation.

14. Performance and Reliability

- Release camera/encoder resources correctly on navigation and lifecycle changes.
- Run filter rendering, thumbnail work and exports off the main thread.
- Use bounded memory for large images and 4K/8K media; stream processing where possible.
- Use thumbnail caching and background export jobs with clear progress.
- Warn before capture/export when storage is insufficient.
- Handle thermal throttling, calls, app interruption, camera contention and permission revocation.
- Never silently reduce output quality; show selected vs actual output profile.

15. Privacy and Permissions

Permission	When	Behavior
Camera	First camera use	Explain before system prompt; required for capture.
Microphone	Video with audio	Allow silent recording where feasible if declined.
Media access	Import/browse external media	Prefer Photo Picker and scoped storage.
Notifications	Only if optional notifications exist	Do not block core camera if declined.
Location metadata	Only if geotagging is offered	Opt-in, configurable and removable on export.

- Avoid broad all-files access unless essential and justified.
- No media uploads unless an optional cloud feature is explicitly introduced and consented to.
- Offer deletion controls for app-created edits and exports.
- Publish an accurate privacy policy for camera, microphone, media and any analytics.

16. Error States

Scenario	Expected response
Permission denied	Explain why; retry/settings route; don't crash.
Unsupported profile	Hide/disable invalid choice with explanation and valid alternative.
Camera busy	Retry safely and explain another app may be using it.
Low storage	Warn before capture/export; offer lower profile or cleanup guidance.
Thermal throttling	Warn and suggest a supported lower profile; never conceal downgrade.
Export fails/cancels	Keep source intact, report status and offer retry.
Codec unavailable	Offer a compatible format and explain trade-off.
Media deleted externally	Refresh library and remove stale item gracefully.
8K unsupported	Explain hardware limitation and suggest highest supported resolution.
Filter failure	Retain original and offer export without the effect.

17. Suggested App Modules and Data

Module	Responsibility
app	Entry point, navigation, lifecycle and theme
camera	Discovery, preview, capture, lens switching and capabilities
pro-controls	Manual controls, profile matrix and applied-state reporting
filters	Preset catalog, preview pipeline, favorites and edit recipes
gallery	MediaStore, thumbnails, viewer and metadata
export	Processing, profile validation, progress, cancellation and save
settings	Themes, defaults, storage, permissions education and help
core-ui	Pills, buttons, sliders, sheets, icons, typography and spacing
core-media	Metadata, codecs, file operations and error mapping

Entity	Key fields
CaptureProfile	mode, cameraId, resolution, FPS, bitrate, codec, ratio, audio
FilterPreset	id, name, family, thumbnail, parameters, version, favorite
MediaItem	URI, type, date, dimensions, duration, format, size, metadata
EditRecipe	source URI, filter, intensity, crop, adjustments, export profile
ExportJob	id, source, profile, status, progress, destination, error
Preferences	theme, grid, default mode/ratio/format, haptics and quality

18. Acceptance Criteria

- Capture creates a valid file and gallery updates after success.
- Front/rear switching works on available cameras and handles unavailable lenses.
- 100+ distinct filters are categorized, searchable and favoritable.
- Filter intensity and before/after work; original remains intact.
- Manual controls are shown only when supported and reflect actual state where available.
- Resolution/FPS/codec combinations are validated and invalid combinations cannot be selected.
- Recording shows elapsed time and actual profile; interruptions are handled safely.
- Exported files are verified for dimensions, codec and metadata where available.
- Custom dimensions enforce limits and explain cropping/upscaling.
- No UI claims native 4K/8K detail where only upscale is performed.
- Gallery supports view, favorite, share, delete and export.
- Themes remain readable at large font sizes and on small screens.
- No major overflow in portrait/landscape across target devices.
- Local capture remains usable offline.

19. QA Test Matrix

Area	Minimum coverage
Devices	Low/mid/high-end Android; front/rear; devices with and without 4K/manual controls
Display	Small/tall/large phones, portrait, landscape and foldable if supported
Capture	Daylight, indoor, backlit, low light, moving subjects, flash/torch
Video	Multiple supported profiles, audio on/off, long recording, thermal/storage pressure
Export	Formats/codecs, crop, resize, quality, cancellation and failure
Accessibility	TalkBack, large fonts, contrast, reduced motion, target size
Lifecycle	Background/foreground, camera contention, permission revocation, rotation
Media	Large library, missing/deleted files, unsupported imports and corrupted sources

20. Implementation Roadmap

Phase	Deliverables
1. Foundation	Theme, navigation, permission education, capability discovery, preview and photo capture
2. Photo experience	Lens switch, grid, flash, EV, aspect ratios, formats and settings
3. Film looks	Filter engine, 100+ presets, live preview, intensity and favorites
4. Pro photo	Supported manual controls, locks and RAW where available
5. Video	Standard recording, valid profile matrix, Pro Video and cinematic guides
6. Gallery/export	Viewer, metadata, crop, custom dimensions, codecs and export jobs
7. Polish	Themes, adaptive layout, accessibility, performance, error states and device QA

21. Release Checklist

- Every visible control performs a real action and has tested states.
- Unsupported hardware features are hidden or explained.
- 100+ presets are visually distinct, categorized and legally appropriate.
- Circular app icon works with Android adaptive icon masking.
- Privacy policy and permission rationale are ready.
- Export formats and codecs are validated on target devices.
- No unsupported 4K/8K claims in marketing or UI.
- Camera lifecycle, thermal, low-storage and export tests pass.
- Accessibility and responsive layouts are checked.
- No demo-only buttons, fake progress or placeholder capture/export.

22. Copy-Ready Build Brief

Build GODCAM as a polished native Android camera application. Use a bold Apple-inspired design language with warm cream backgrounds, subtle mint accents, deep forest-charcoal text, simple vector icons, pill-shaped mode selectors and actions, circular icon controls, rounded sheets and clean typography. Keep the live camera preview central and make the entire UI adaptive to phone size, orientation, safe areas and system font scale.

Implement real photo capture and video recording. Include front/rear switching, flash/torch, grid, exposure compensation, timer, aspect ratios, photo size/format options, Night mode where supported and Pro Photo with supported manual controls. Ship 100+ distinct film-inspired looks across Fuji-inspired, Kodak-inspired, retro, cinematic, monochrome, portrait, nature and creative families. Add preview, intensity, favorites, search, categories and before/after comparison.

Add Standard Video and Pro Video with runtime-supported resolution, FPS, bitrate, codec, focus, exposure, white balance, stabilization and audio controls. Validate every profile against actual device capability. Include cinematic aspect guides and label guide vs export crop. Add a built-in gallery with photos, videos, favorites, viewer, metadata, share, delete and high-quality export.

Create Export Studio with custom photo/video dimensions, numeric width/height plus sliders, aspect-ratio lock, resolution, frame rate, codec/format, bitrate/quality, crop, audio and metadata controls. Save real files, show accurate progress, support cancellation and errors, preserve originals by default, and never promise unsupported 4K/8K. Request permissions contextually, prefer Photo Picker and scoped storage, and keep local capture usable offline. Avoid generic AI-style visuals, clutter, fake features and decorative gradients. Test across supported Android devices.

23. Product Principle

GODCAM should feel like a beautifully crafted camera first and a powerful tool second. The default path must be effortless; advanced controls must be discoverable; every export must be honest about its quality; and every screen must respect the user's display, attention and creative intent.